Unequal Environments, Unequal Health: Associations between Deprivation, Green Cover, and Year 6 Obesity across London MSOAs

Preparation

- https://github.com/Levine-I/CASA0006_project
- Number of words: 1498
- Runtime: 1:31 Minutes (*System: Google Colab, Memory ~13 GB, CPU Intel(R) Xeon(R))*
- Coding environment: Python 3.12
- License: this notebook is made available under the Creative Commons Attribution license.
- Additional library *[libraries not included in SDS Docker or not used in this module]*:
  - geopandas(1.1.3): handling MSOA boundary geometries and spatial joins
  - statsmodels(0.14.6): OLS regression with HC3 robust standard errors and VIF diagnostics
  - libpysal(4.14.1): construction of spatial weights matrices
  - esda(2.9.0): calculation of Moran's I for regression residuals

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Introduction

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Nowadays, childhood obesity remains a significant public health issue in England, especially among older primary school children. In 2023/24, Year 6 children obesity prevalence in England was 22.1%, with prevalence ranging from 29.2% in the most deprived areas to 13.0% in the least deprived. London is also among the regions with the highest Year 6 obesity prevalence, making it a ideal place to investigate spatial health inequalities (NHS England, 2024; Office for Health Improvement and Disparities, 2024).

Previous research suggests that childhood obesity has a clear spatial and neighbourhood dimension. Across England, obesity prevalence shows positive spatial dependence and is associated with socioeconomic disadvantage (Sun et al., 2020). Evidence also suggests that socioeconomic inequalities in child obesity may vary across ethnic groups (Goisis et al., 2019), while neighbourhood greenspace may also matter (Mears et al., 2020). Therefore, this project examines how Year 6 obesity prevalence across London MSOAs is associated with area deprivation, socio-demographic characteristics, and neighbourhood green cover.

Research questions

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To what extent is Year 6 obesity prevalence across London MSOAs associated with area deprivation, socio-demographic composition, and neighbourhood green cover?

Data

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Data Introduction

This project uses five datasets at MSOA level.

The dependent variable is taken from the Prevalence of Childhood Obesity, Borough, Ward and MSOA dataset from the London Datastore, which reports childhood weight outcomes for Reception and Year 6 pupils at multiple spatial scales, including MSOA.

Neighbourhood deprivation is measured using the English Indices of Deprivation 2019, and socio-demographic characteristics are drawn from the 2021 Census Bulk Data Download, which includes census tables for MSOA such as ethnicity, housing tenure, education level, population density and household composition.

The environmental variable is based on the Green Cover 2024 dataset, which provides estimates of green cover in London and related spatial layers. Finally, London's Statistical GIS Boundary Files are used to support spatial referencing and mapping at MSOA level.

Data Processing

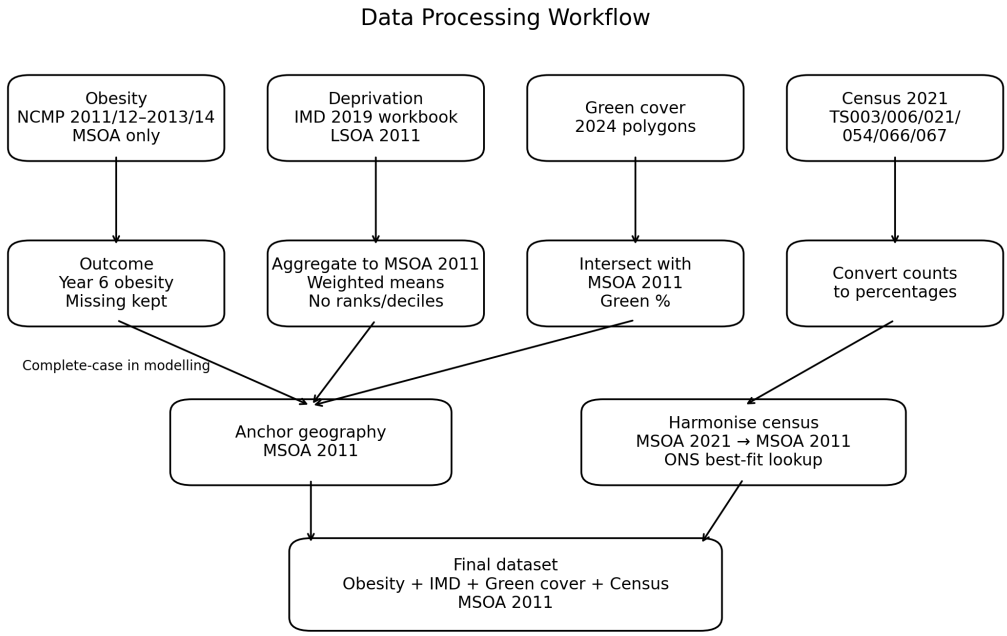
For child obesity prevalence data, extract from the worksheet 2011-12 to 2013-14, and only retain at MSOA level. A small number of missing values which may due to factors such as poor response rates and selective opt-outs, would handled through complete-case analysis during modelling (Information Centre for Health and Social Care, n.d.; Allison, 2001).

Deprivation data are using the IMD 2019, IDACI and IDAOPI, and Population figures sheets from the London IMD worksheet, and aggregated from 2011 LSOA to MSOA level using population weighted means.

Neighbourhood green cover was processed from the 2024 green cover polygon tiles, and spatial intersecting with London MSOA boundaries. Then calculate the total green cover area within each MSOA and converted into a percentage of the total MSOA area.

Socio-demographic covariates included population density, ethnic composition, housing tenure, unemployment and economic inactivity, qualifications, and household structure. Original counts are converted into percentages, and ethnic composition is represented by pct_non_white to reduce model multicollinearity.

Because the datasets used different geography boundary, Census 2021 and Green Cover data were harmonised to MSOA 2011 using official ONS lookups (Best fit & exact fit). The final dataset combined obesity, IMD, green cover, and census covariates on MSOA 2011 geography.



Variables used in the analysis

Variable	Type	Description	Notes
Year 6 obesity prevalence (`year6_obese_pct`)	Numeric	Percentage of Year 6 children classified as obese at MSOA level. Used as the dependent variable in the analysis.	Main outcome variable.
Green cover percentage (`green_cover_pct`)	Numeric	Percentage of land area classified as green cover within each MSOA. Used as the main exposure variable.	Main exposure of interest.
Index of Multiple Deprivation score (`imd_score_weighted`)	Numeric	Population-weighted IMD 2019 score at MSOA level, representing overall area deprivation.	Main deprivation measure used in the core models.
Income Deprivation Affecting Children Index score (`idaci_score_weighted`)	Numeric	Child-population-weighted IDACI score at MSOA level, reflecting deprivation affecting children.	Used in sensitivity analysis rather than the main model due to overlap with IMD.

Variable	Type	Description	Notes
Population density (`population_density_per_sqkm`)	Numeric	Number of residents per square kilometre in each MSOA.	Included as a socio-demographic control.
Non-white population percentage (`pct_non_white`)	Numeric	Percentage of residents identifying as non-white in each MSOA.	Broad socio-demographic covariate used to reduce multicollinearity.
Social rented housing percentage (`pct_social_rented`)	Numeric	Percentage of households living in socially rented accommodation in each MSOA.	Socio-demographic covariate; may also reflect housing disadvantage.
No qualifications percentage (`pct_no_qualifications`)	Numeric	Percentage of residents with no formal qualifications in each MSOA.	Socio-demographic covariate.
Unemployment percentage (`pct_unemployed_16plus`)	Numeric	Percentage of residents aged 16+ who are unemployed in each MSOA.	Socio-demographic covariate.
Lone-parent household percentage (`pct_lone_parent_households`)	Numeric	Percentage of households headed by a lone parent in each MSOA.	Socio-demographic covariate.
MSOA code (`MSOA_code`)	Categorical / Identifier	Unique code identifying each MSOA.	Used for joining only.
MSOA name (`MSOA_name`)	Categorical / Identifier	Name of each MSOA.	Used for reporting and mapping only.
Local authority district name (`LAD_name`)	Categorical / Identifier	Name of the local authority district containing the MSOA.	
Exact fit match count (`exactfit_match_count`)	Numeric	Validation count for spatial matching.	Used for data checking only.

Methodology

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Analytical Strategy

This project uses a cross-sectional, MSOA-level design to examine how London Year 6 obesity prevalence is associated with neighbourhood green cover, area deprivation, and selected socio-demographic characteristics. The dependent variable is year6_obese_pct. The main exposure element is green_cover_pct. Area deprivation is primarily measured using the population weighted IMD 2019 score (imd_score_weighted), while IDACI is for sensitivity analysis because of it is conceptually close with IMD.

Exploratory Analysis

EDA begins with descriptive and spatial exploratory analysis, using histograms, summary statistics, choropleth maps, scatterplots, and a correlation matrix to examine variable distributions, spatial patterning, and bivariate relationships across London MSOAs. This is intended to identify broad inequalities, motivate the later modelling strategy, and flag potential issues such as skewness and collinearity.

Regression, Diagnostics, and Spatial checks

The main inferential approach is hierarchical multivariable linear regression with HC3 robust standard errors.

Model 1 includes deprivation only. Model 2 adds selected socio-demographic covariates, including population density, ethnic composition, housing tenure, educational disadvantage, unemployment, and lone-parent household prevalence. Model 3 then adds green cover.

$$\text{Model 1: } y_i = \beta_0 + \beta_1 \text{IMD}_i + \varepsilon_i$$

$$\text{Model 2: } y_i = \beta_0 + \beta_1 \text{IMD}_i + \beta' \mathbf{X}_i + \varepsilon_i$$

$$\text{Model 3: } y_i = \beta_0 + \beta_1 \text{IMD}_i + \beta' \mathbf{X}_i + \beta_2 \text{GreenCover}_i + \varepsilon_i$$

(X_i includes selected socio-demographic covariates)

I assess multicollinearity using the correlation matrix and variance inflation factors, and exclude cases with missing values on model variables using complete-case analysis. None of the VIF values exceed commonly used rule-of-thumb thresholds for severe multicollinearity, often taken to be around 5 to 10, so the variables were retained, while overlapping disadvantage indicators were interpreted with care (Kim, 2019; O'Brien, 2007).

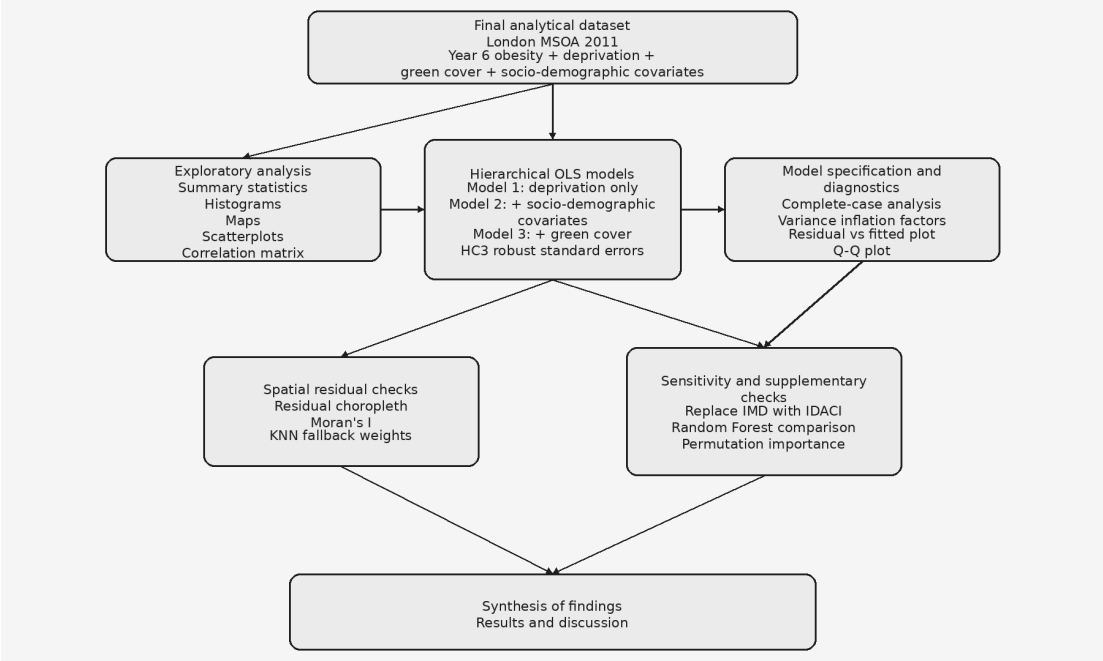
$$VIF_j = \frac{1}{1 - R_j^2}$$

Then residual spatial patterning in the fully adjusted model is examined using a residual choropleth and Moran's I. Because Queen contiguity weights produced isolates for this geometry, Moran's I was calculated using a centroid-based K-nearest-neighbours weights matrix (k = 8) as an alternative specification, this is

consistent with Anselin (2020), who notes that isolates can be problematic for spatial analysis.

Sensitivity Analysis and Comparison Model

The sensitivity analysis aim to test whether the findings depend on how deprivation is measured by replacing imd_score_weighted with idaci_score_weighted. Besides, random forest is included as a supplementary comparison model to assess if variable importance rankings are consistent under a non-linear specification.



Exploratory Analysis

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Loading data

In [39]:
```
from pathlib import Path
import geopandas as gpd
from IPython.display import display
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
from statsmodels.stats.outliers_influence import variance_inflation_facto
import warnings
warnings.filterwarnings("ignore", category=FutureWarning)

#plotting style and resolution
sns.set_theme(style="whitegrid", context="talk")
plt.rcParams["figure.dpi"] = 120
plt.rcParams["savefig.dpi"] = 300
```

```
#path from the github
data_path = "https://raw.githubusercontent.com/Levine-l/CASA0006_project/
boundary_path = "https://raw.githubusercontent.com/Levine-l/CASA0006_proj

#variables to be used in the regression analysis
eda_vars = [
    "MSOA_code",
    "MSOA_name",
    "LAD_name",
    "year6_obese_pct",
    "imd_score_weighted",
    "green_cover_pct",
    "population_density_per_sqkm",
    "pct_non_white",
    "pct_social_rented",
    "pct_no_qualifications",
    "pct_unemployed_16plus",
    "pct_lone_parent_households",
]

#load the main analysis dataset from GitHub
analysis = pd.read_csv(data_path)
#make sure MSOA codes are strings and stripped of whitespace for accurate
analysis["MSOA_code"] = analysis["MSOA_code"].astype(str).str.strip()

#load the London MSOA 2011 boundary
boundary = gpd.read_file(boundary_path)[["MSOA11CD", "MSOA11NM", "LAD11NM
#rename boundary columns to match the main dataset identifiers
boundary = boundary.rename(
    columns={
        "MSOA11CD": "MSOA_code",
        "MSOA11NM": "MSOA_name_boundary",
        "LAD11NM": "LAD_name_boundary",
    }
)
boundary["MSOA_code"] = boundary["MSOA_code"].astype(str).str.strip()

#validate that all required core variables are present in the loaded data
missing_vars = [col for col in eda_vars if col not in analysis.columns]
if missing_vars:
    raise KeyError(f"Missing expected columns in harmonised dataset: {mis

#create a subset of data for eda
eda = analysis[eda_vars].copy()
#merge analysis data with spatial boundaries for mapping
eda_map = boundary.merge(eda, on="MSOA_code", how="left")

#print summary of loaded data
print(f"Analysis data: {analysis.shape[0]} rows x {analysis.shape[1]} col
print(f"Boundary data: {boundary.shape[0]} MSOAs")
print(f"Records with obesity data after spatial merge: {eda_map['year6_ob
```

```
Analysis data: 983 rows x 61 columns
Boundary data: 983 MSOAs
Records with obesity data after spatial merge: 959
```

1. Data Overview

In [40]:
```
#calculate and display missing values for each variable
missingness = pd.DataFrame({
```

```
    "variable": eda_vars,
    "missing_n": [eda[col].isna().sum() for col in eda_vars],
})
missingness["missing_pct"] = (missingness["missing_n"] / len(eda) * 100).
missingness = missingness.sort_values(["missing_pct", "variable"], ascend

#show the missingness summary table
display(missingness)

#define the subset of numeric variables
summary_vars = [
    "year6_obese_pct",
    "imd_score_weighted",
    "green_cover_pct",
    "population_density_per_sqkm",
    "pct_non_white",
    "pct_social_rented",
    "pct_no_qualifications",
    "pct_unemployed_16plus",
    "pct_lone_parent_households",
]

#descriptive statistics for the chosen variables
summary_table = eda[summary_vars].describe().T[["count", "mean", "std", "
summary_table = summary_table.round(2)

#display summary statistics table
display(summary_table)
```

	variable	missing_n	missing_pct
0	year6_obese_pct	24	2.44
1	LAD_name	0	0.00
2	MSOA_code	0	0.00
3	MSOA_name	0	0.00
4	green_cover_pct	0	0.00
5	imd_score_weighted	0	0.00
6	pct_lone_parent_households	0	0.00
7	pct_no_qualifications	0	0.00
8	pct_non_white	0	0.00
9	pct_social_rented	0	0.00
10	pct_unemployed_16plus	0	0.00
11	population_density_per_sqkm	0	0.00

	count	mean	std	min	25%	50%	
year6_obese_pct	959.0	21.83	5.46	5.63	18.46	22.58	
imd_score_weighted	983.0	21.50	9.34	4.04	13.77	21.02	
green_cover_pct	983.0	39.36	16.16	3.18	27.13	37.81	
population_density_per_sqkm	983.0	8846.13	4966.55	302.30	5019.20	7896.30	12
pct_non_white	983.0	45.01	17.51	6.62	30.86	43.61	
pct_social_rented	983.0	22.66	15.00	1.32	10.43	19.54	
pct_no_qualifications	983.0	16.18	5.60	4.67	11.69	15.86	
pct_unemployed_16plus	983.0	4.77	1.30	1.70	3.71	4.77	
pct_lone_parent_households	983.0	13.33	4.82	3.56	9.74	12.61	

Only Y6 obesity data has a few missing values.

2. Histogram of Y6 Obesity Prevalence

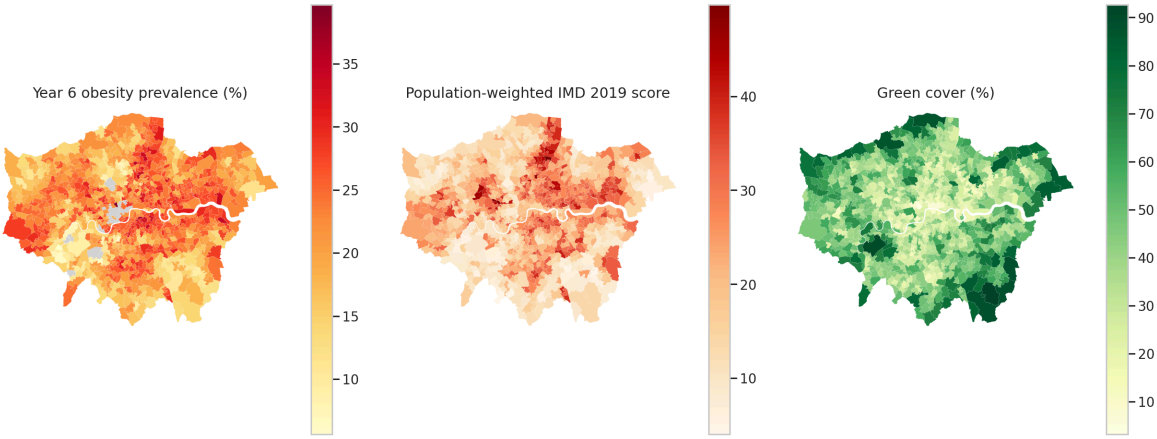
In [41]:
```
#figure size for the plot
fig, ax = plt.subplots(figsize=(8, 5))

#histogram where x is the Year 6 obesity percentage and y is the number o
#with a kernel density estimate curve plot
sns.histplot(data=eda, x="year6_obese_pct", bins=25, kde=True, color="#bc

ax.set_title("Distribution of Year 6 obesity prevalence across London MSO
ax.set_xlabel("Year 6 obesity prevalence (%)")
ax.set_ylabel("Number of MSOAs")

#adjust layout
fig.tight_layout()

# Display the plot in the notebook output
plt.show()
```

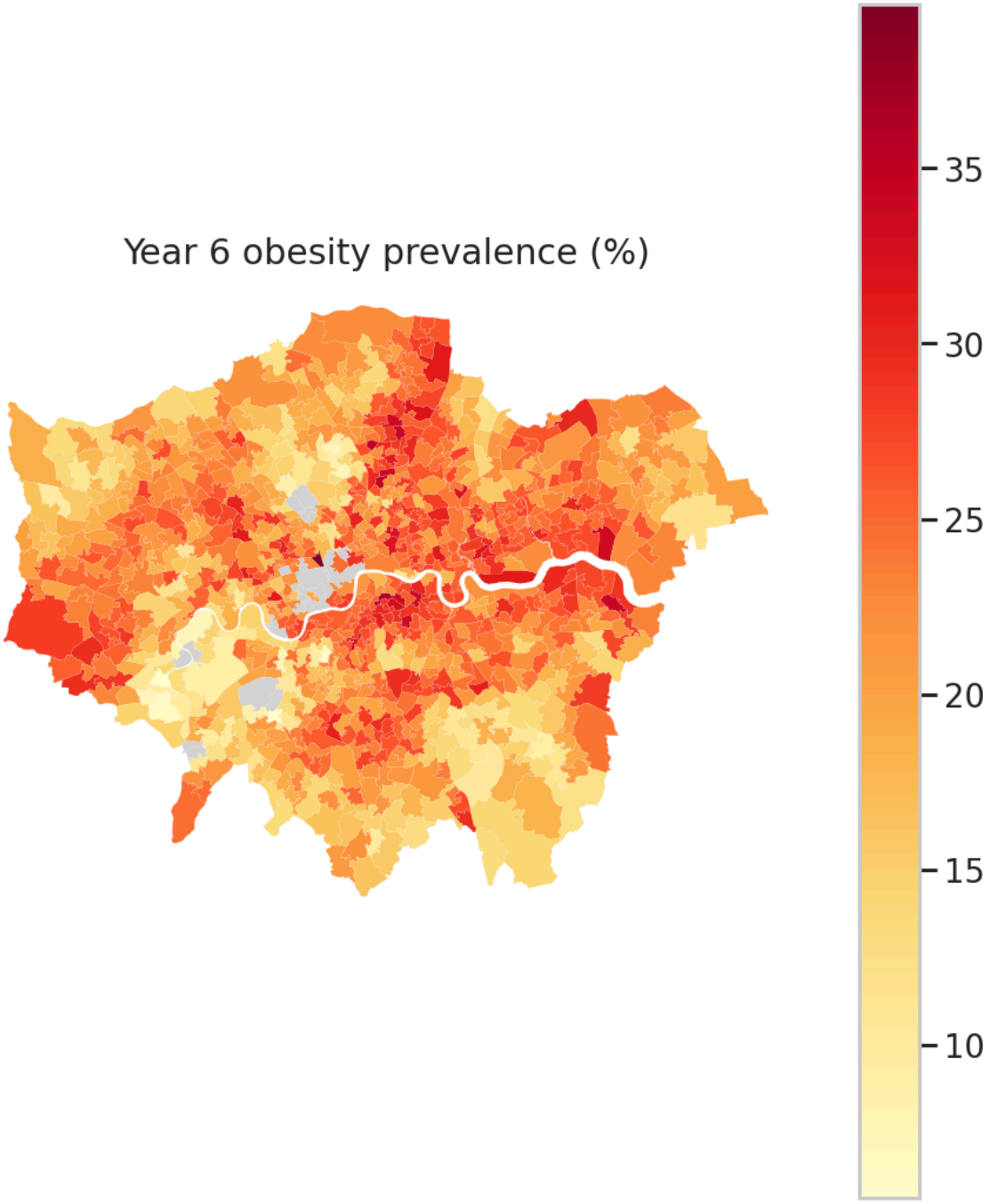
The distribution of Year 6 obesity prevalence is concentrated between 15% and 30%, with a modal peak at approximately 24%. While a small number of MSOAs exhibit relatively low or high prevalence, but there is no obvious skew in the overall distribution.

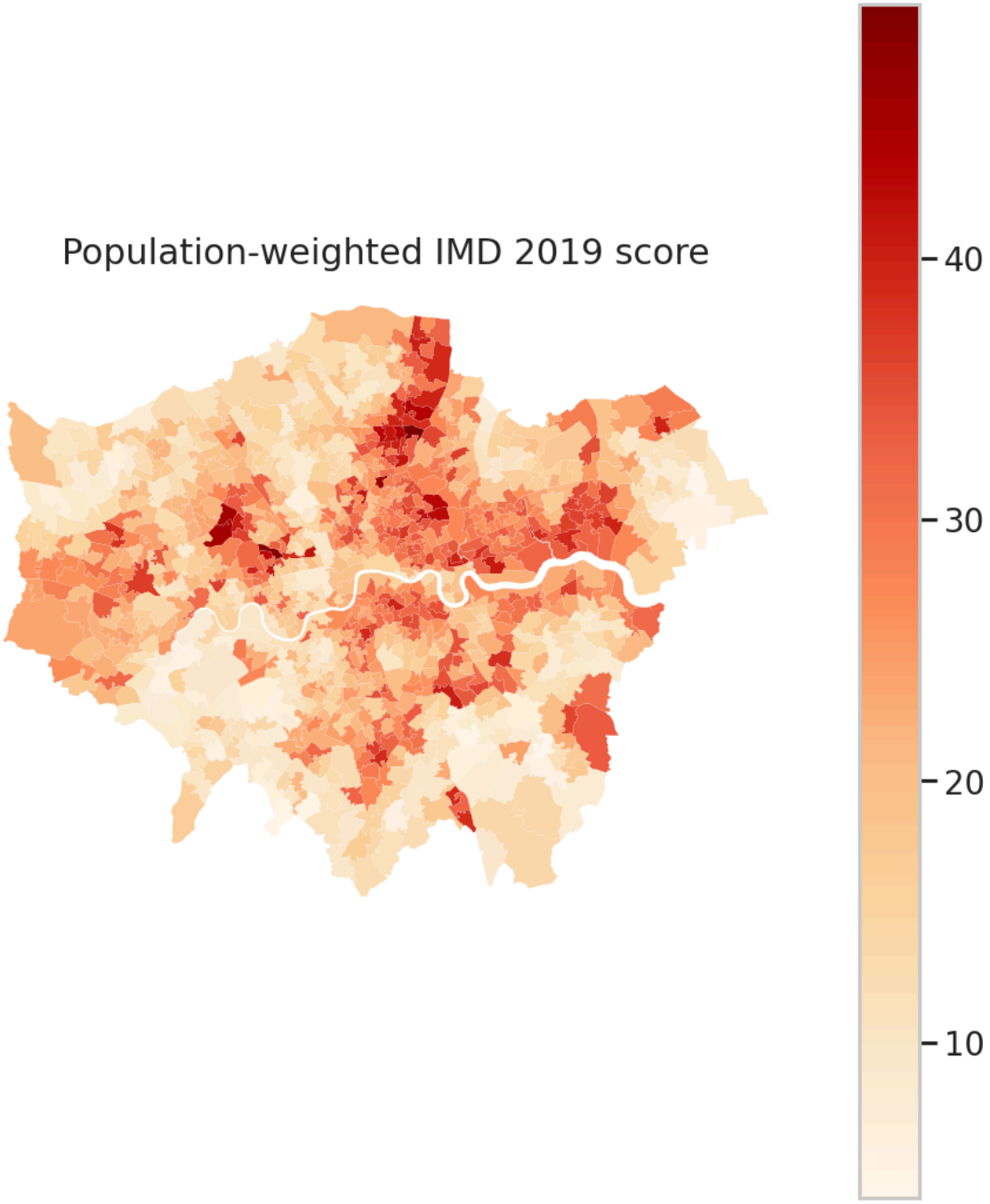
3. Choropleth Map

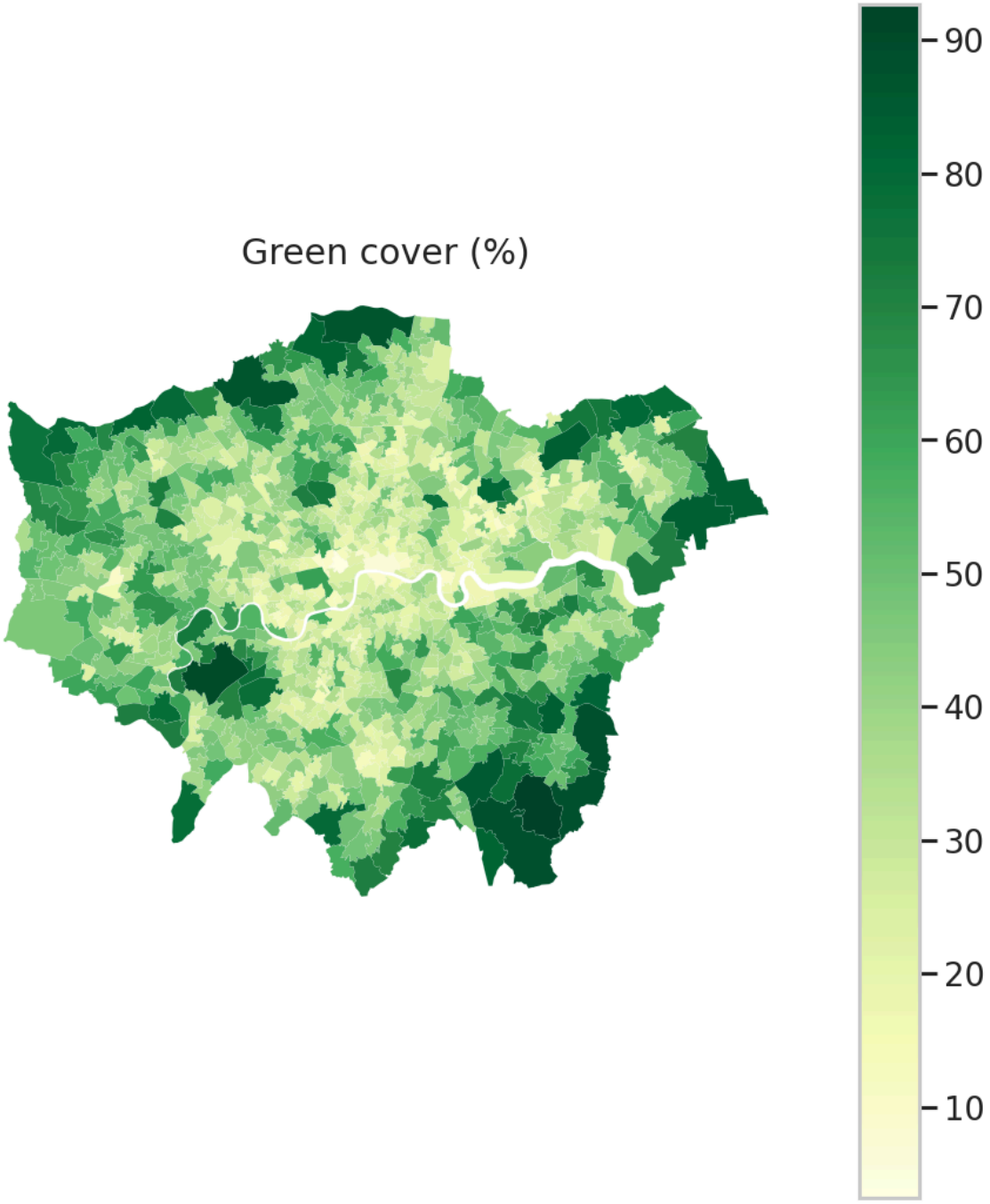
In [42]:
```
#define the map settings
map_specs = [
    ("year6_obese_pct", "Year 6 obesity prevalence (%)", "YlOrRd"),
    ("imd_score_weighted", "Population-weighted IMD 2019 score", "OrRd"),
    ("green_cover_pct", "Green cover (%)", "YlGn"),
]

#combined 3 plot figure for overview
fig, axes = plt.subplots(1, 3, figsize=(21, 8))

#plot the choropleth map
for ax, (column, title, cmap) in zip(axes, map_specs):
    eda_map.plot(
        column=column,
        cmap=cmap,
        linewidth=0.05,
        edgecolor="white",
        legend=True,
        missing_kwds={"color": "lightgrey", "label": "Missing"}, #dealing
        ax=ax,
    )
    ax.set_title(title)
    ax.set_axis_off() #hide x y coordinates

fig.tight_layout()
plt.show()

#create larger individual maps
for column, title, cmap in map_specs:
    fig, ax = plt.subplots(figsize=(8, 9))
    eda_map.plot(
```

```
        column=column,
        cmap=cmap,
        linewidth=0.05,
        edgecolor="white",
        legend=True,
        missing_kwds={"color": "lightgrey", "label": "Missing"},
        ax=ax,
    )
    ax.set_title(title)
    ax.set_axis_off()
    fig.tight_layout()
    plt.show()
```




Year 6 obesity prevalence (%)
35
30
25
20
15
10


Population-weighted IMD 2019 score
40
30
20
10

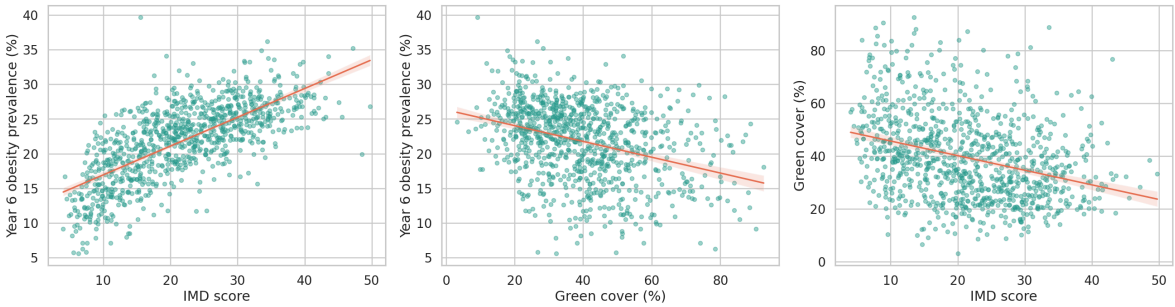


The maps show that obesity and deprivation got a similar spatial pattern, while green cover is more concentrated in outer London.

4. Key Scatterplots

In [43]:
```
#define the scatterplot pairs
scatter_specs = [
    ("imd_score_weighted", "year6_obese_pct", "IMD score", "Year 6 obesit
    ("green_cover_pct", "year6_obese_pct", "Green cover (%)", "Year 6 obe
    ("imd_score_weighted", "green_cover_pct", "IMD score", "Green cover (
]

#a multi-plot figure for key bivariate relationships
fig, axes = plt.subplots(1, 3, figsize=(22, 6))

for ax, (x_var, y_var, x_label, y_label) in zip(axes, scatter_specs):
    plot_df = eda[[x_var, y_var]].dropna().copy()#drop missing values
```

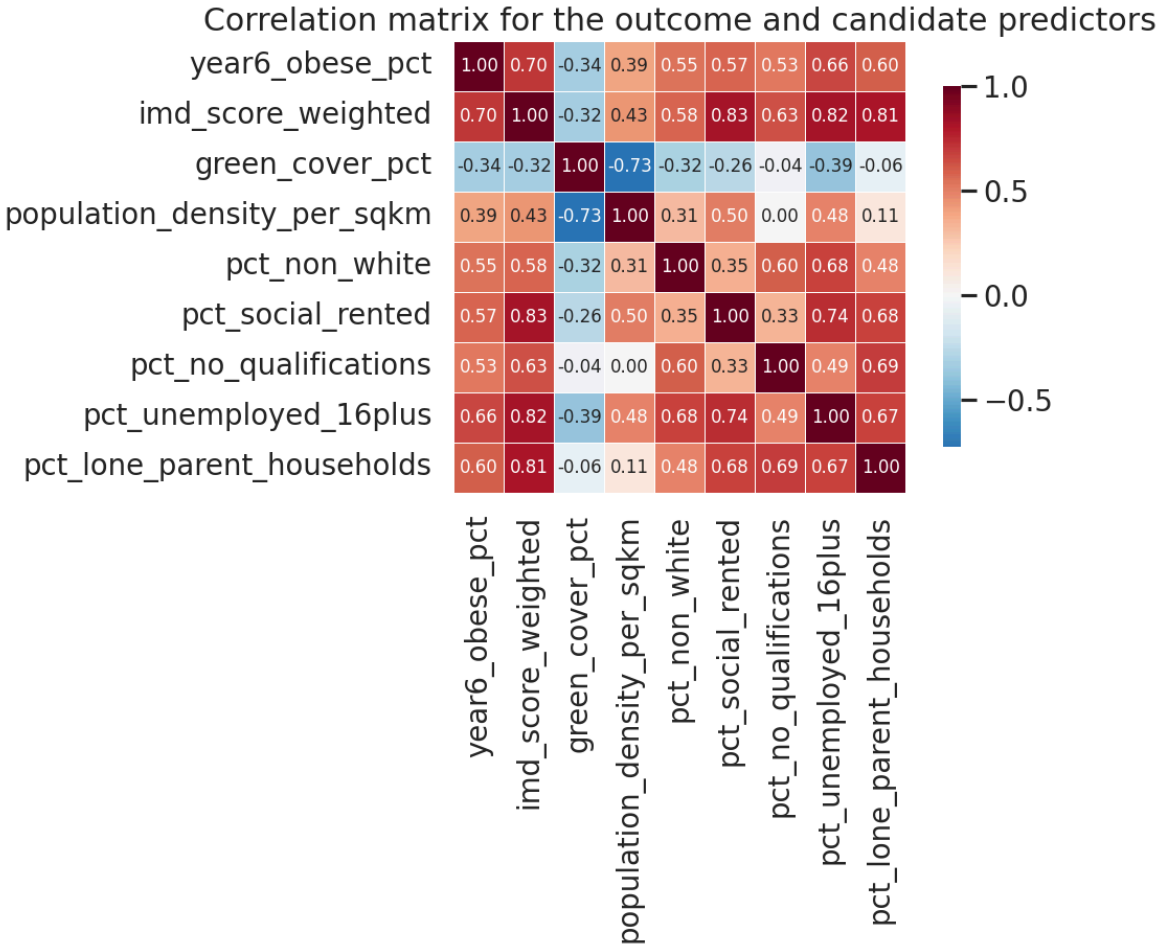
```
    #seaborn regplot to show scatter points and fitted linear regression
    sns.regplot(
        data=plot_df,
        x=x_var,
        y=y_var,
        scatter_kws={"alpha": 0.45, "s": 28, "color": "#2a9d8f"}, #point'
        line_kws={"color": "#e76f51", "lw": 2}, #regression line appearan
        ax=ax,
    )
    ax.set_xlabel(x_label)
    ax.set_ylabel(y_label)

#adjust layout
fig.tight_layout()
plt.show()
```



5. Correlation Heatmap

In [44]:
```
#variables for the correlation analysis
heatmap_vars = [
    "year6_obese_pct",
    "imd_score_weighted",
    "green_cover_pct",
    "population_density_per_sqkm",
    "pct_non_white",
    "pct_social_rented",
    "pct_no_qualifications",
    "pct_unemployed_16plus",
    "pct_lone_parent_households",
]

#compute Pearson correlation matrix
corr_df = eda[heatmap_vars].copy()
corr_matrix = corr_df.corr(numeric_only=True)

#the heatmap visualization
fig, ax = plt.subplots(figsize=(10, 8))
sns.heatmap(
    corr_matrix,
    cmap="RdBu_r", #red-blue color map
    center=0,
    annot=True, #print the correlation coefficients in each cell
    annot_kws={"size": 10},#aadjust the size of numbers
    fmt=".2f",#format coefficients to 2 decimal places
    square=True,#ensure cells are square shape
    linewidths=0.5,
    cbar_kws={"shrink": 0.8},
    ax=ax,
```

```
)

ax.set_title("Correlation matrix for the outcome and candidate predictors
fig.tight_layout()

plt.show()

#a ranked list of correlations with obesity
corr_with_outcome = corr_matrix["year6_obese_pct"].sort_values(ascending=
display(corr_with_outcome.to_frame(name="correlation_with_year6_obese_pct
```



	correlation_with_year6_obese_pct
year6_obese_pct	1.000000
imd_score_weighted	0.704459
pct_unemployed_16plus	0.664553
pct_lone_parent_households	0.598388
pct_social_rented	0.574534
pct_non_white	0.546697
pct_no_qualifications	0.525397
population_density_per_sqkm	0.394047
green_cover_pct	-0.336009

In the scatterplots and the correlation matrix, Year 6 obesity prevalence is positively related to IMD and negatively related to green cover. Several disadvantage related covariates are also positively correlated with obesity and deprivation.

Results and discussion

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In [45]:
```
#group variables
identifier_vars = ["MSOA_code", "MSOA_name", "LAD_name"] #labels
outcome = "year6_obese_pct"#dependent variable
deprivation_var = "imd_score_weighted"#primary predictor
green_var = "green_cover_pct" #exposure variable

#additional socio-demographic variables
sociodemographic_vars = [
    "population_density_per_sqkm",
    "pct_non_white",
    "pct_social_rented",
    "pct_no_qualifications",
    "pct_unemployed_16plus",
    "pct_lone_parent_households",
]

#combine all core variables into a single list for data subset
model_core_vars = identifier_vars + [outcome, deprivation_var, green_var]

#alternative deprivation measure for robustness check
sensitivity_vars = ["idaci_score_weighted"]

#define the components for hierarchical modeling
model_1_vars = [deprivation_var] #baseline which is deprivation only
model_2_additions = sociodemographic_vars.copy() #add socio-demographics
model_3_additions = [green_var] #add green cover
```

1. Build the Modelling Dataset

In [46]:
```
#all unique variables required for the primary and sensitivity models
required_vars = model_core_vars + sensitivity_vars
required_vars = list(dict.fromkeys(required_vars)) #remove duplicates and

#check if any required variables are missing from the input dataframe
missing_vars = [col for col in required_vars if col not in analysis.colum
if missing_vars:
    raise KeyError(f"Missing expected columns in harmonised dataset: {mis

#a subset dataframe containing only the necessary modeling columns
model_data = analysis[required_vars].copy()

#calculate missingness each variable to evaluate data quality for the reg
missingness = pd.DataFrame({
    "variable": required_vars,
    "missing_n": [model_data[col].isna().sum() for col in required_vars],
})
missingness["missing_pct"] = (missingness["missing_n" ] / len(model_data)
display(missingness.sort_values(["missing_pct", "variable"], ascending=[F
```

```
#identify variables needed for regression models (models 1-3)
core_model_vars = [outcome] + model_1_vars + model_2_additions + model_3_

#drop any rows with missing values in the outcome or core predictors (com
complete_case = model_data.dropna(subset=core_model_vars).copy()
complete_case = complete_case.reset_index(drop=True)

#print summary
print(f"Rows in full harmonised dataset: {len(model_data)}")
print(f"Rows in complete-case regression dataset: {len(complete_case)}")
print(f"Rows dropped for main regression models: {len(model_data) - len(c
```

	variable	missing_n	missing_pct
0	year6_obese_pct	24	2.44
1	LAD_name	0	0.00
2	MSOA_code	0	0.00
3	MSOA_name	0	0.00
4	green_cover_pct	0	0.00
5	idaci_score_weighted	0	0.00
6	imd_score_weighted	0	0.00
7	pct_lone_parent_households	0	0.00
8	pct_no_qualifications	0	0.00
9	pct_non_white	0	0.00
10	pct_social_rented	0	0.00
11	pct_unemployed_16plus	0	0.00
12	population_density_per_sqkm	0	0.00

```
Rows in full harmonised dataset: 983
Rows in complete-case regression dataset: 959
Rows dropped for main regression models: 24
```

2. Fit Hierarchical Linear Regression Models

In [47]:
```
#model 1 exploring the association with deprivation only
formula_1 = f"{outcome} ~ {deprivation_var}"
#model 2 adding socio-demographic control variables
formula_2 = f"{outcome} ~ {deprivation_var} + {' + '.join(sociodemographi
#model 3 full model adding the green cover
formula_3 = f"{formula_2} + {green_var}"

#fit the models using ordinary least squares
model_1 = smf.ols(formula=formula_1, data=complete_case).fit(cov_type="HC
model_2 = smf.ols(formula=formula_2, data=complete_case).fit(cov_type="HC
model_3 = smf.ols(formula=formula_3, data=complete_case).fit(cov_type="HC

#print the summaries
#print(model_1.summary())
#print(model_2.summary())
#print(model_3.summary())
```

```
#store the full summaries for checking, but keep the final notebook outpu
model_summaries = {
    "Model 1": model_1.summary(),
    "Model 2": model_2.summary(),
    "Model 3": model_3.summary(),
}
print("Models fitted successfully. Full statsmodels summaries are retaine
```

```
Models fitted successfully. Full statsmodels summaries are retained in `mo
del_summaries` for checking but are not displayed here.
```

In [48]:

```
#define a function to extract and format key regression statistics
def tidy_model(model, model_name):
    conf = model.conf_int()
    out = pd.DataFrame({
        "term": model.params.index,
        "coef": model.params.values,
        "std_err": model.bse.values,
        "p_value": model.pvalues.values,
        "ci_low": conf[0].values,
        "ci_high": conf[1].values,
        "model": model_name,
    })
    return out

#combine the results from all three models into a single table for compar
tidy_results = pd.concat([
    tidy_model(model_1, "Model 1"),
    tidy_model(model_2, "Model 2"),
    tidy_model(model_3, "Model 3"),
], ignore_index=True)

#display the combined results
display(tidy_results.round(4))
```

	term	coef	std_err	p_value	ci_low	ci_high	model
0	Intercept	12.7946	0.3503	0.0000	12.1081	13.4810	Model 1
1	imd_score_weighted	0.4150	0.0140	0.0000	0.3875	0.4426	Model 1
2	Intercept	7.9851	0.6787	0.0000	6.6548	9.3155	Model 2
3	imd_score_weighted	0.1484	0.0382	0.0001	0.0734	0.2233	Model 2
4	population_density_per_sqkm	0.0002	0.0000	0.0000	0.0001	0.0002	Model 2
5	pct_non_white	0.0322	0.0100	0.0014	0.0125	0.0519	Model 2
6	pct_social_rented	0.0032	0.0172	0.8508	-0.0305	0.0370	Model 2
7	pct_no_qualifications	0.1529	0.0367	0.0000	0.0810	0.2248	Model 2
8	pct_unemployed_16plus	0.6649	0.2215	0.0027	0.2309	1.0990	Model 2
9	pct_lone_parent_households	0.1372	0.0519	0.0082	0.0354	0.2390	Model 2
10	Intercept	9.3023	1.1096	0.0000	7.1275	11.4770	Model 3
11	imd_score_weighted	0.1404	0.0384	0.0003	0.0652	0.2156	Model 3
12	population_density_per_sqkm	0.0001	0.0000	0.0089	0.0000	0.0002	Model 3
13	pct_non_white	0.0318	0.0101	0.0016	0.0121	0.0515	Model 3
14	pct_social_rented	0.0123	0.0183	0.5006	-0.0236	0.0482	Model 3
15	pct_no_qualifications	0.1569	0.0368	0.0000	0.0848	0.2289	Model 3
16	pct_unemployed_16plus	0.6287	0.2221	0.0047	0.1933	1.0640	Model 3
17	pct_lone_parent_households	0.1349	0.0518	0.0092	0.0334	0.2364	Model 3
18	green_cover_pct	-0.0194	0.0122	0.1120	-0.0434	0.0045	Model 3

In [49]:
```
#build a summary table of model
model_fit = pd.DataFrame([
    {
        "model": "Model 1",
        "formula": formula_1,
        "n": int(model_1.nobs),
```

```
        "r_squared": model_1.rsquared,
        "adj_r_squared": model_1.rsquared_adj,
        "aic": model_1.aic,
        "bic": model_1.bic,
    },
    {
        "model": "Model 2",
        "formula": formula_2,
        "n": int(model_2.nobs),
        "r_squared": model_2.rsquared,
        "adj_r_squared": model_2.rsquared_adj,
        "aic": model_2.aic,
        "bic": model_2.bic,
    },
    {
        "model": "Model 3",
        "formula": formula_3,
        "n": int(model_3.nobs),
        "r_squared": model_3.rsquared,
        "adj_r_squared": model_3.rsquared_adj,
        "aic": model_3.aic,
        "bic": model_3.bic,
    },
]).round(4)

#display the fit summary
display(model_fit)
```

	model	formula	n	r_squared	adj_r_squared	aic	bic
0	Model 1	year6_obese_pct ~ imd_score_weighted	959	0.4963	0.4957	5323.4460	5333.1778
1	Model 2	year6_obese_pct ~ imd_score_weighted + populat...	959	0.5567	0.5535	5212.7946	5251.7218
2	Model 3	year6_obese_pct ~ imd_score_weighted + populat...	959	0.5581	0.5544	5211.8715	5255.6645

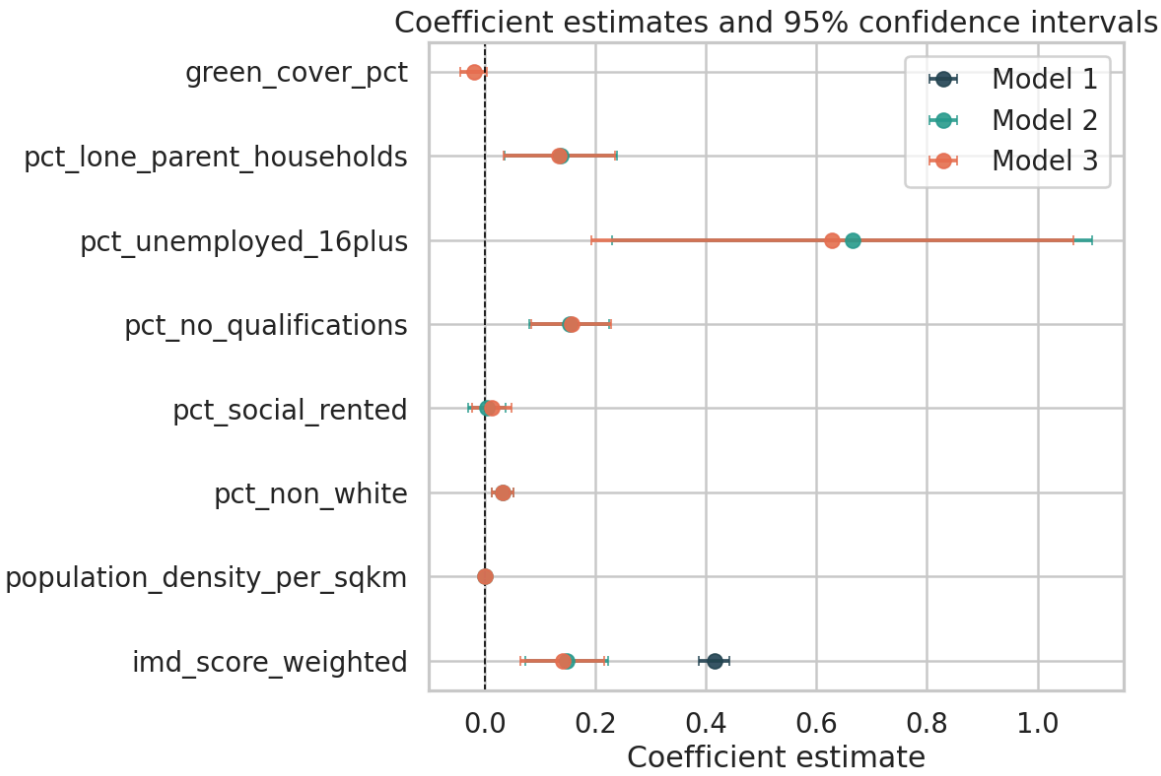
The regression results of model 1 indicates that deprivation alone can explains a substantial ratio of variation in Year 6 obesity prevalence.

Model 2 improves fit after socio-demographic covariates are added, while Model 3 shows only limited further improvement after green cover is included, means deprivation and socio-demographic context can contribute more explanatory power than green cover.

3. Coefficient Comparison Plot

In [50]:
```
#excluding the intercept
plot_df = tidy_results.loc[tidy_results["term"] != "Intercept"].copy()
#sort terms from top to bottom
plot_df["term"] = pd.Categorical(plot_df["term"], categories=list(reverse

fig, ax = plt.subplots(figsize=(10, 7))
```

```
palette = {"Model 1": "#264653", "Model 2": "#2a9d8f", "Model 3": "#e76f5

#plot point estimates and 95% confidence intervals for each hierarchical
for model_name, sub in plot_df.groupby("model"):
    ax.errorbar(
        x=sub["coef"],
        y=sub["term"],
        xerr=[sub["coef"] - sub["ci_low"], sub["ci_high" ] - sub["coef"]]
        fmt="o",
        capsize=3,
        label=model_name,
        color=palette[model_name],
        alpha=0.9,
    )

#add a reference line at zero to identify statistically significant predi
ax.axvline(0, color="black", linewidth=1, linestyle="--")
ax.set_title("Coefficient estimates and 95% confidence intervals")
ax.set_xlabel("Coefficient estimate")
ax.set_ylabel("")
ax.legend(frameon=True)
fig.tight_layout()
plt.show()
```



The coefficient plot shows that IMD remains positively related with obesity after adjustment, although its effect is reduced.

No qualifications, lone-parent households and especially unemployment remain positively associated with obesity, whereas green cover retains only a weak negative coefficient with limited evidence of an independent effect.

4. Multicollinearity Diagnostics

In [51]:
```
#calculate VIF to check for multicollinearity among predictors
vif_vars = model_1_vars + model_2_additions + model_3_additions
vif_df = complete_case[vif_vars].copy()
vif_df = sm.add_constant(vif_df)

vif_table = pd.DataFrame({
    "variable": vif_df.columns,
    "vif": [variance_inflation_factor(vif_df.values, i) for i in range(vi
})
#table remove constant and sort by vif values
vif_table = vif_table.loc[vif_table["variable"] != "const"].sort_values("

display(vif_table.round(3))
```

	variable	vif
0	imd_score_weighted	8.166
1	pct_social_rented	5.344
2	pct_unemployed_16plus	4.505
3	pct_lone_parent_households	4.244
4	population_density_per_sqkm	3.398
5	pct_no_qualifications	2.933
6	pct_non_white	2.555
7	green_cover_pct	2.438

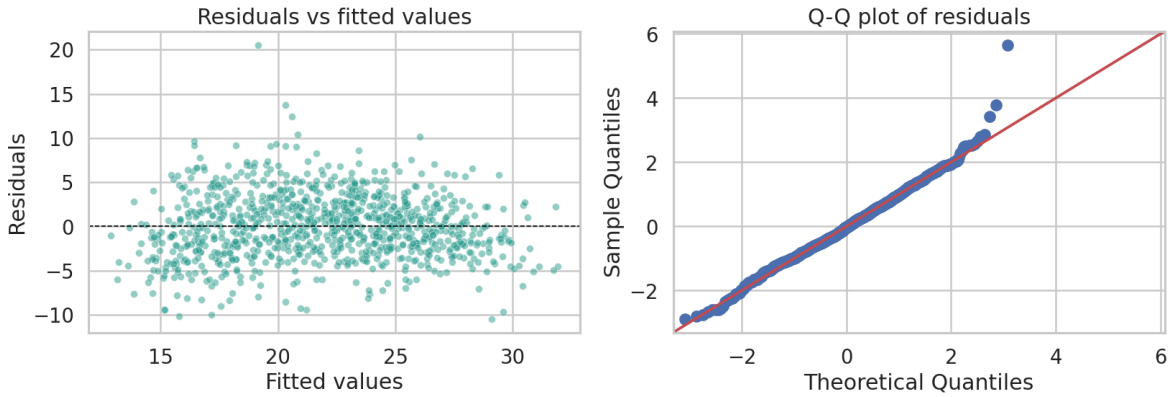
We can see some overlap among several disadvantage related predictors from the results of variance inflation factor, particularly the IMD score and measures such as social renting, unemployment, and lone-parent households.

However, none of the VIF values exceed conventional thresholds for severe multicollinearity, so the model was kept.

5. Residual Diagnostics for the Preferred Model

In [52]:
```
#perform residual analysis on model 3
preferred_model = model_3
diagnostic_df = complete_case.copy()
diagnostic_df["fitted"] = preferred_model.fittedvalues
diagnostic_df["residuals"] = preferred_model.resid
diagnostic_df["std_residuals"] = preferred_model.get_influence().resid_st

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

#check constant variance on left plot
sns.scatterplot(data=diagnostic_df, x="fitted", y="residuals", alpha=0.5,
axes[0].axhline(0, color="black", linestyle="--", linewidth=1)
axes[0].set_title("Residuals vs fitted values")
axes[0].set_xlabel("Fitted values")
axes[0].set_ylabel("Residuals")

#check for normality of residuals on right plot
```

```
sm.qqplot(diagnostic_df["residuals"], line="45", fit=True, ax=axes[1])
axes[1].set_title("Q-Q plot of residuals")

fig.tight_layout()
plt.show()
```

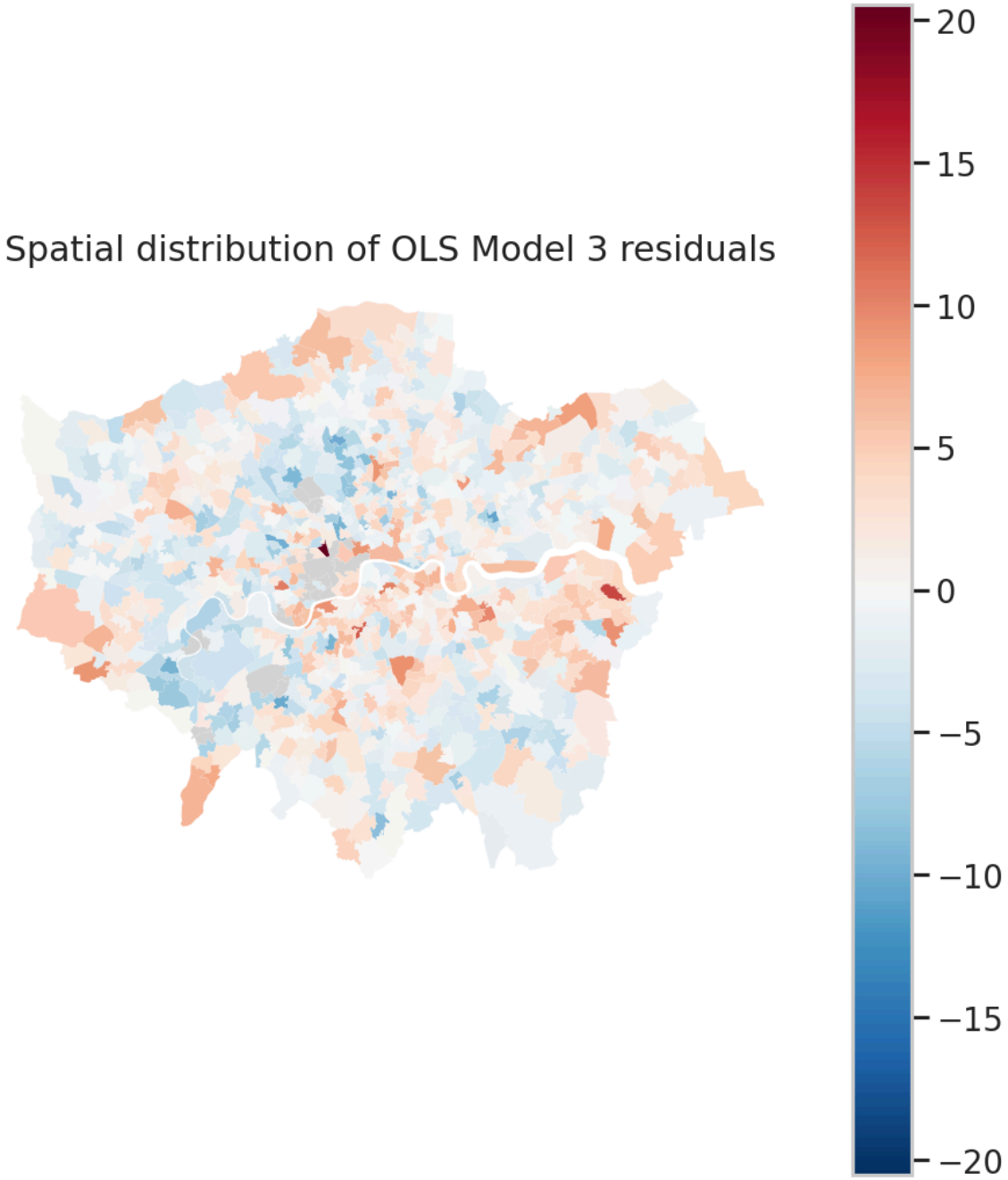


Residuals are broadly centred around zero by the plotting, with no strong non-linear pattern which supports the use of a linear specification.

Although there are some deviations in the upper tail of the Q–Q plot, but residuals follow the normal reference line well in the centre of the distribution, so only a small number of outlying.

5.1 Spatial Pattern of Model 3 Residuals

In [53]:
```
#map the residuals from model 3 to check whether unexplained variation st
residual_map = boundary.merge(
    diagnostic_df[["MSOA_code", "residuals"]],
    on="MSOA_code",
    how="left",
)

max_abs_residual = np.nanmax(np.abs(residual_map["residuals"]))

fig, ax = plt.subplots(figsize=(8, 9))
residual_map.plot(
    column="residuals",
    cmap="RdBu_r",
    vmin=-max_abs_residual,
    vmax=max_abs_residual,
    linewidth=0.05,
    edgecolor="white",
    legend=True,
    missing_kwds={"color": "lightgrey", "label": "Missing"},
    ax=ax,
)
ax.set_title("Spatial distribution of OLS Model 3 residuals")
ax.set_axis_off()
fig.tight_layout()
plt.show()
```



Although much of the remaining error is relatively small and spatially dispersed, some localised clusters of positive and negative residuals are still visible.

In [54]:
```
#if the required spatial statistics packages are available, estimate Mora
try:
    import warnings
    from libpysal.weights import KNN, Queen
    from esda.moran import Moran

    residual_geo = residual_map.dropna(subset=["residuals"]).copy().reset

    #use a projected CRS for distance-based fallback weights if needed
    if residual_geo.crs is not None and residual_geo.crs.to_epsg() != 277
        residual_geo = residual_geo.to_crs(27700)

    #first try polygon contiguity, which is the most natural choice for a
    with warnings.catch_warnings():
        warnings.simplefilter("ignore")
        queen_weights = Queen.from_dataframe(residual_geo, use_index=Fals
```

```
    if len(getattr(queen_weights, "islands", [])) == len(residual_geo):
        #if contiguity fails because all areas are treated as islands, fa
        residual_centroids = residual_geo.copy()
        residual_centroids["geometry"] = residual_centroids.geometry.cent
        weights = KNN.from_dataframe(residual_centroids, k=8)
        weight_type = "KNN (k=8)"
        island_count = len(getattr(weights, "islands", []))
        print("Queen contiguity produced only islands, so Moran's I uses
    else:
        weights = queen_weights
        weight_type = "Queen contiguity"
        island_count = len(getattr(weights, "islands", []))

    weights.transform = "r"
    moran_model3 = Moran(residual_geo["residuals"].values, weights, permu

    moran_table = pd.DataFrame([
        {
            "statistic": "Moran's I",
            "weight_type": weight_type,
            "n_observations": len(residual_geo),
            "n_islands": island_count,
            "value": moran_model3.I,
            "expected_value": moran_model3.EI,
            "p_value_permutation": moran_model3.p_sim,
            "z_score_permutation": moran_model3.z_sim,
        }
    ]).round(4)

    display(moran_table)

except ModuleNotFoundError:
    print("Moran's I was skipped because libpysal/esda is not installed i
except Exception as exc:
    print("Moran's I could not be computed in the current environment.")
    print(exc)
```

```
Queen contiguity produced only islands, so Moran's I uses KNN weights inst
ead.
```

	statistic	weight_type	n_observations	n_islands	value	expected_value	p_valu
0	Moran's I	KNN (k=8)	959	0	0.1761	-0.001	

Moran's I also proof that there's positive spatial autocorrelation in the residuals.

Because Queen contiguity produced only islands for this geometry, the test was calculated using a centroid-based K-nearest neighbours weights matrix (k=8).

In particular, the model under-predicts or over-predicts obesity prevalence in some areas, means that some neighbourhood-level influences may remain unobserved.

6. Sensitivity Analysis: Replace IMD with IDACI

In [55]:
```
#a sensitivity analysis that testing if findings are robust using a child
sensitivity_vars = [outcome, "idaci_score_weighted"] + sociodemographic_v
```

```
sensitivity_data = model_data.dropna(subset=sensitivity_vars).copy().rese

#sensitivity formula
formula_sensitivity = f"{outcome} ~ idaci_score_weighted + {' + '.join(so

#fit the model and compare results with the IMD model
model_sensitivity = smf.ols(formula=formula_sensitivity, data=sensitivity
print(model_sensitivity.summary())

#tidy up for easier comparison
sensitivity_tidy = tidy_model(model_sensitivity, "Sensitivity model")
display(sensitivity_tidy.round(4))
```

```
                            OLS Regression Results
==============================================================================
====
Dep. Variable:        year6_obese_pct   R-squared:
0.586
Model:                            OLS   Adj. R-squared:
0.583
Method:                 Least Squares   F-statistic:                     1
58.1
Date:                Fri, 10 Apr 2026   Prob (F-statistic):          8.19e
-169
Time:                        15:01:32   Log-Likelihood:                -25
65.5
No. Observations:                 959   AIC:                             5
149.
Df Residuals:                     950   BIC:                             5
193.
Df Model:                           8
Covariance Type:                  HC3
==============================================================================
======================
                                 coef    std err          z      P>|z|
[0.025      0.975]
------------------------------------------------------------------------------
---------------------
Intercept                      9.2613      1.093      8.477      0.000
7.120      11.403
idaci_score_weighted          30.9514      3.648      8.485      0.000
23.802      38.101
population_density_per_sqkm  9.677e-05   4.69e-05      2.064      0.039
4.89e-06       0.000
pct_non_white                  0.0399      0.010      4.098      0.000
0.021       0.059
pct_social_rented             -0.0331      0.017     -1.896      0.058
-0.067       0.001
pct_no_qualifications          0.1735      0.031      5.621      0.000
0.113       0.234
pct_unemployed_16plus          0.4853      0.212      2.285      0.022
0.069       0.902
pct_lone_parent_households     0.0518      0.051      1.018      0.309
-0.048       0.151
green_cover_pct               -0.0182      0.012     -1.486      0.137
-0.042       0.006
==============================================================================
====
Omnibus:                       52.932   Durbin-Watson:
1.730
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               9
5.798
Skew:                           0.393   Prob(JB):                     1.58
e-21
Kurtosis:                       4.334   Cond. No.                     3.08
e+05
==============================================================================
====

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
[2] The condition number is large, 3.08e+05. This might indicate that ther
```

```
e are
strong multicollinearity or other numerical problems.
```

	term	coef	std_err	p_value	ci_low	ci_high	mod
0	Intercept	9.2613	1.0925	0.0000	7.1200	11.4027	Sensitivi mod
1	idaci_score_weighted	30.9514	3.6478	0.0000	23.8020	38.1009	Sensitivi mod
2	population_density_per_sqkm	0.0001	0.0000	0.0390	0.0000	0.0002	Sensitivi mod
3	pct_non_white	0.0399	0.0097	0.0000	0.0208	0.0589	Sensitivi mod
4	pct_social_rented	-0.0331	0.0175	0.0580	-0.0673	0.0011	Sensitivi mod
5	pct_no_qualifications	0.1735	0.0309	0.0000	0.1130	0.2340	Sensitivi mod
6	pct_unemployed_16plus	0.4853	0.2124	0.0223	0.0690	0.9016	Sensitivi mod
7	pct_lone_parent_households	0.0518	0.0509	0.3086	-0.0479	0.1515	Sensitivi mod
8	green_cover_pct	-0.0182	0.0123	0.1373	-0.0423	0.0058	Sensitivi mod

In sensitivity analysis, replacing IMD with IDACI produce similar results, which suggest that the main findings are not sensitive to the deprivation measure used.

7. Comparison Model: Random Forest

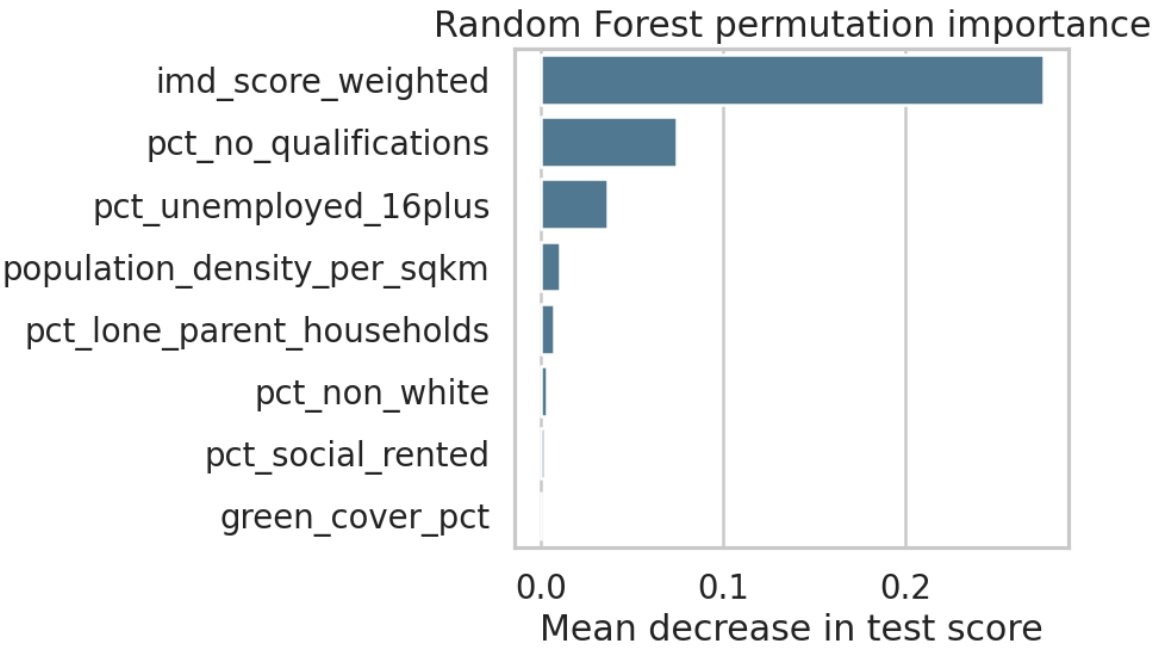
To assess whether the relative importance of predictors is generally consistent under a nonlinear setting, random forest regression model is estimated as an assist comparison model.

In [56]:
```
from sklearn.ensemble import RandomForestRegressor
from sklearn.inspection import permutation_importance
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split

rf_vars = model_1_vars + model_2_additions + model_3_additions
rf_data = complete_case[[outcome] + rf_vars].dropna().copy()

X = rf_data[rf_vars]
y = rf_data[outcome]

#split training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25,

rf = RandomForestRegressor(
    n_estimators=500,
    max_depth=None,
    min_samples_leaf=3,
    random_state=42,
```

```
    n_jobs=-1,
)
rf.fit(X_train, y_train)

#evaluate model accuracy
y_pred = rf.predict(X_test)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))#calculate root mean sq

rf_metrics = pd.DataFrame([
    {
        "model": "Random Forest",
        "test_rmse": rmse,
        "test_r_squared": r2_score(y_test, y_pred),
    }
]).round(4)
display(rf_metrics)

#calculate permutation importance
perm = permutation_importance(rf, X_test, y_test, n_repeats=25, random_st
importance_df = pd.DataFrame({
    "variable": X.columns,
    "importance_mean": perm.importances_mean,
    "importance_std": perm.importances_std,
}).sort_values("importance_mean", ascending=False).reset_index(drop=True)

display(importance_df.round(4))

#show variable importance rank
fig, ax = plt.subplots(figsize=(8, 5))
sns.barplot(data=importance_df, y="variable", x="importance_mean", color=
ax.set_title("Random Forest permutation importance")
ax.set_xlabel("Mean decrease in test score")
ax.set_ylabel("")
fig.tight_layout()
plt.show()
```

	model	test_rmse	test_r_squared
0	Random Forest	3.6864	0.4966

	variable	importance_mean	importance_std
0	imd_score_weighted	0.2760	0.0450
1	pct_no_qualifications	0.0746	0.0281
2	pct_unemployed_16plus	0.0368	0.0123
3	population_density_per_sqkm	0.0103	0.0088
4	pct_lone_parent_households	0.0073	0.0138
5	pct_non_white	0.0033	0.0171
6	pct_social_rented	0.0021	0.0053
7	green_cover_pct	-0.0002	0.0105



The random forest model which has better flexibility did not improve predictive performance, and again identified deprivation related variables as the most important predictors rather than green cover.

Limitations

This project uses cross-sectional data from multiple sources that are not perfectly contemporaneous, so it identifies associations rather than causal effects. Also, MSOA level findings may not apply to individuals.

In addition, Moran's I shows some spatial dependence remains unexplained, suggesting omitted neighbourhood processes or residual spatial structure. Given the word limit and the focus on interpretable baseline association models, spatial regression is left for future work.

Furthermore, the use of `pct_non_white` as a broad measure of ethnic composition is a little rough and does not capture the significant heterogeneity in obesity risks and socio-economic outcomes across different specific ethnic groups.

Conclusion

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In conclusion, Year 6 obesity prevalence is strongly associated with area deprivation, which this relationship is consistent across both the OLS and random forest model. Several indicators of neighbourhood disadvantage, particularly educational disadvantage and unemployment, also retain positive associations with obesity prevalence in the adjusted models.

Although greener areas tend to have lower obesity prevalence, but after adjusting for deprivation and socio-demographic context it does not remain robust. This suggests

that the apparent relationship between greener neighbourhoods and lower obesity prevalence is likely to reflect wider neighbourhood conditions.

Overall, the results indicate that deprivation related neighbourhood disadvantage plays a more important role in explaining spatial variation in Year 6 obesity prevalence across London MSOAs, and policies should focus more on wider socioeconomic disadvantage than on green cover alone.

References

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